

Операционные системы

Основы интерфейса взаимодействия пользователя с системой Unix на уровне командной строки

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2026-09-02

Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

- 1 Определить имя и путь домашнего каталога
- 2 Изучить команду `ls`.
- 3 Выполнить действия с каталогами.
- 4 Получить дополнительные сведения при помощи справки по командам.
- 5 Изучить команду `history`.

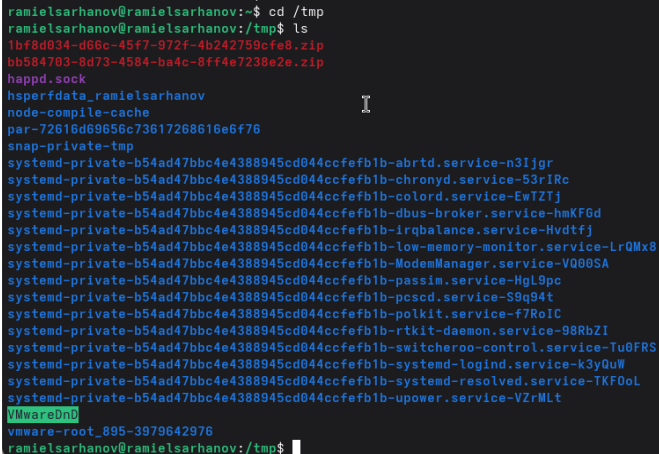
2. Процесс выполнения лабораторной работы

A terminal window with a dark background. The title bar at the top shows a plus icon on the left and the text 'ramielsarhanov@ramielsarhanov:~' on the right. Below the title bar, the terminal displays a series of commands and their outputs. The prompt 'ramielsarhanov@ramielsarhanov:~\$' is shown in green. The first command is 'cd', which is followed by a new prompt. The second command is 'pwd', which outputs '/home/ramielsarhanov' in white. The third prompt 'ramielsarhanov@ramielsarhanov:~\$' is followed by a white cursor block.

```
ramielsarhanov@ramielsarhanov:~$ cd
ramielsarhanov@ramielsarhanov:~$ pwd
/home/ramielsarhanov
ramielsarhanov@ramielsarhanov:~$ █
```

Рисунок 1: Путь к домашнему каталогу

Опции команды ls



```
ramielsarhanov@ramielsarhanov:~$ cd /tmp
ramielsarhanov@ramielsarhanov:/tmp$ ls
1bf8d034-d66c-45f7-972f-4b242759cfe8.zip
bb584703-8d73-4584-ba4c-8ff4e7238e2e.zip
hapd.sock
hsperfdata_ramielsarhanov
node-compile-cache
par-72616d69656c73617268616e6f76
snap-private-tmp
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-abrt.service-n3Ijgr
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-chrony.service-53rIRc
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-colord.service-EwTZTj
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-dbus-broker.service-hmKF6d
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-irqbalance.service-Hvdtfj
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-low-memory-monitor.service-LrQMx8
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-ModemManager.service-VQ00SA
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-passim.service-Hgl9pc
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-pcsd.service-S9q94t
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-polkit.service-f7RoIC
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-rtkit-daemon.service-98RbZI
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-switcheroo-control.service-Tu0FRS
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-systemd-logind.service-k3yQuW
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-systemd-resolved.service-TKF0oL
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-upower.service-VZrMLt
VMwareDnD
vmware-root_895-3979642976
ramielsarhanov@ramielsarhanov:/tmp$
```

Рисунок 2: Команда ls


```
ramielsarhanov@ramielsarhanov:/tmp$ ls -a
.
..
1bf8d034-d66c-45f7-972f-4b242759cfe8.zip
bb584703-8d73-4584-ba4c-8ff4e7238e2e.zip
.font-unix
happd.sock
hsperfdata_ramielsarhanov
.ICE-unix
node-compile-cache
par-72616d69656c73617268616e66f76
snap-private-tmp
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-abrt.service-n3ljgr
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-chronyd.service-53rIRc
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-colord.service-EwTZTj
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-dbus-broker.service-hmKFGd
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-irqbalance.service-Hvdtfj
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-low-memory-monitor.service-LrQMx8
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-ModemManager.service-VQ00SA
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-passim.service-HgL9pc
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-pcsd.service-S9q94t
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-polkit.service-f7RoIC
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-rtkit-daemon.service-98RbZI
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-switcheroo-control.service-Tu0FRS
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-systemd-logind.service-k3yQuW
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-systemd-resolved.service-TKF0oL
systemd-private-b54ad47bbc4e4388945cd044ccfefb1b-upower.service-VZrMLt
VMwareDnD
vmware-root_895-3979642976
.X0-lock
.X11-unix
```

Рисунок 3: Команда ls -a

Опции команды ls

```
ramielsarhanov@ramielsarhanov:/tmp$ ls -l
total 7712
-rw-r--r--. 1 ramielsarhanov ramielsarhanov 3947343 Sep  2 15:09 1bf8d034-d66c-45f7-972f-4b242759cfe8.zip
-rw-r--r--. 1 ramielsarhanov ramielsarhanov 3947343 Sep  2 15:12 bb584703-8d73-4584-ba4c-8ff4e7238e2e.zip
srw-rw-rw-. 1 root root 0 Sep  2 15:07 happd.sock
drwxr-xr-x. 2 ramielsarhanov ramielsarhanov 40 Sep  2 15:26 hspdfdata_ramielsarhanov
drwxr-xr-x. 3 ramielsarhanov ramielsarhanov 60 Sep  2 15:35 node-compile-cache
drwx-----. 3 ramielsarhanov ramielsarhanov 60 Sep  2 15:23 par-72616d69656c73617268616e6f76
drwx-----. 2 root root 40 Sep  2 15:07 snap-private-tmp
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-abrt.d.service-n3Ijgr
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-chronyd.service-53rIRc
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-colord.service-EwTZTj
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-dbus-broker.service-hmKFGd
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-irqbalance.service-Hvdtfj
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-low-memory-monitor.service-LrQMx8
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-ModemManager.service-VQ00SA
drwx-----. 3 root root 60 Sep  2 15:08 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-passim.service-HgL9pc
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-pcsd.service-S9q94t
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-polkit.service-f7RoIC
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-rtkit-daemon.service-98RbZI
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
b1b-switcheroo-control.service-Tu0FRS
drwx-----. 3 root root 60 Sep  2 15:07 systemd-private-b54ad47bbc4e4388945cd044ccfef
```

Рисунок 4: Команда ls -l

Опции команды ls

```
ramielsarhanov@ramielsarhanov:/tmp$ ls -f
.
..
node-compile-cache
hspanferdata_ramielsarhanov
par-72616d69656c73617268616e6f76
VMwareDnD
bb584703-8d73-4584-ba4c-8ff4e7238e2e.zip
1bf8d034-d66c-45f7-972f-4b242759cfe8.zip
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-passim.service-HgI9pc
.X1-lock
.X0-lock
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-pcsd.service-S9q94t
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-colord.service-EwTzTj
happd.sock
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-ModemManager.service-VQ00SA
vmware-root_895-3979642976
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-systemd-logind.service-k3yQuW
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-upower.service-VZrMLt
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-switcheroo-control.service-Tu0FRS
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-rtkit-daemon.service-98RbZI
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-polkit.service-f7RoIC
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-low-memory-monitor.service-LrQMx8
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-chrond.service-53rIRc
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-irqbalance.service-Hvdtfj
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-abrt.service-n3Ijgr
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-dbus-broker.service-hmKF6d
systemd-private-b54ad47bbc4e4388945cd044ccfebf1b-systemd-resolved.service-TKF0oL
.font-unix
.XIM-unix
.ICE-unix
.X11-unix
snap-private-tmp
ramielsarhanov@ramielsarhanov:/tmp$
```

Рисунок 5: Команда ls -f

```
ramielsarhanov@ramielsarhanov:/tmp$ cd /var/spool/  
ramielsarhanov@ramielsarhanov:/var/spool$ ls -l  
total 0  
drwxr-x--x. 1 root abrt 24476 Aug 31 13:37 abrt  
drwx-----. 1 abrt abrt    0 Dec  4 2025 abrt-upload  
drwxr-xr-x. 1 root root   66 Oct 23 2025 anacron  
drwx-----. 1 root root   18 Oct 23 2025 at  
drwx-----. 1 root root    0 Aug  6 2025 cron  
drwx--x---. 1 root lp      6 Dec 12 2025 cups  
drwxr-xr-x. 1 root root    0 Jul 30 2025 lpd  
drwxrwxr-x. 1 root mail  244 Sep  2 15:06 mail  
drwxr-xr-x. 1 root root    0 Jan 27 2026 plymouth  
ramielsarhanov@ramielsarhanov:/var/spool$
```

Рисунок 6: Каталог /var/spool

```
ramielsarhanov@ramielsarhanov:/var/spool$ cd
ramielsarhanov@ramielsarhanov:~$ ls
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work
ramielsarhanov@ramielsarhanov:~$ ls -al
total 24
drwx-----. 1 ramielsarhanov ramielsarhanov 370 Sep  2 15:45 .
drwxr-xr-x. 1 root            root          244 Sep  2 15:06 ..
-rw-----. 1 ramielsarhanov ramielsarhanov 2668 Sep  2 15:46 .bash_history
-rw-r--r--. 1 ramielsarhanov ramielsarhanov  18 Jul 23 2025 .bash_logout
-rw-r--r--. 1 ramielsarhanov ramielsarhanov 144 Jul 23 2025 .bash_profile
-rw-r--r--. 1 ramielsarhanov ramielsarhanov 684 Sep  2 15:35 .bashrc
drwx-----. 1 ramielsarhanov ramielsarhanov 484 Sep  2 15:38 .cache
drwx-----. 1 ramielsarhanov ramielsarhanov 360 Sep  2 15:38 .config
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov   0 Sep  2 15:08 Desktop
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov   0 Sep  2 15:08 Documents
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov   0 Sep  2 15:08 Downloads
-rw-r--r--. 1 ramielsarhanov ramielsarhanov 334 Nov 14 2025 .emacs
-rw-r--r--. 1 ramielsarhanov ramielsarhanov 239 Sep  2 15:45 .gitconfig
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov  74 Sep  2 15:41 git-extended
drwx-----. 1 ramielsarhanov ramielsarhanov 136 Sep  2 15:17 .gnupg
drwx-----. 1 ramielsarhanov ramielsarhanov  20 Sep  2 15:08 .local
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov  34 Oct 23 2025 .mozilla
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov   0 Sep  2 15:08 Music
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov   0 Sep  2 15:08 Pictures
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov   0 Sep  2 15:08 Public
drwx-----. 1 ramielsarhanov ramielsarhanov 132 Sep  2 15:19 .ssh
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov   0 Sep  2 15:08 Templates
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov  18 Sep  2 15:22 .texlive2025
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov   0 Sep  2 15:08 Videos
drwxr-xr-x. 1 ramielsarhanov ramielsarhanov  10 Sep  2 15:19 work
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 7: Файлы в домашнем каталоге

```
ramielsarhanov@ramielsarhanov:~$  
ramielsarhanov@ramielsarhanov:~$ mkdir newdir  
ramielsarhanov@ramielsarhanov:~$ mkdir newdir/morefun  
ramielsarhanov@ramielsarhanov:~$ mkdir letters memos misk  
ramielsarhanov@ramielsarhanov:~$ ls  
Desktop  Downloads  letters  misk  newdir  Public  Videos  
Documents  git-extended  memos  Music  Pictures  Templates  work  
ramielsarhanov@ramielsarhanov:~$ rm letters/ memos/ misk/  
rm: cannot remove 'letters/': Is a directory  
rm: cannot remove 'memos/': Is a directory  
rm: cannot remove 'misk/': Is a directory  
ramielsarhanov@ramielsarhanov:~$ rm -r letters/ memos/ misk/  
ramielsarhanov@ramielsarhanov:~$ rm -r newdir/  
ramielsarhanov@ramielsarhanov:~$ ls  
Desktop  Documents  Downloads  git-extended  Music  Pictures  Public  Templates  Videos  work  
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 8: Действия с каталогами

```
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_extensions/yamadh  
arma/minted-quarto':  
_extension.yml minted-quarto.lua  
  
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/image':  
solvay.jpg  
  
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources':  
csl tex  
  
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/csl':  
gost-r-7-0-5-2008-numeric.csl  
  
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/report/_resources/tex':  
preamble.tex  
  
./work/study/2025-2026/Операционные системы/2026-1--study--os-intro/template/report/scripts':  
image-report mpv-shot  
ramielsarhanov@ramielsarhanov:~$ ls -t  
git-extended work Desktop Documents Downloads Music Pictures Public Templates Videos  
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 9: Команда ls -R и ls -t

```
ramielsarhanov@ramielsarhanov:~$ help cd
cd: cd [-L|[-P [-e]]] [-@] [dir]
    Change the shell working directory.

    Change the current directory to DIR.  The default DIR is the value of the
    HOME shell variable.  If DIR is "-", it is converted to $OLDPWD.

    The variable CDPATH defines the search path for the directory containing
    DIR.  Alternative directory names in CDPATH are separated by a colon (:).
    A null directory name is the same as the current directory.  If DIR begins
    with a slash (/), then CDPATH is not used.

    If the directory is not found, and the shell option `cdable_vars' is set,
    the word is assumed to be a variable name.  If that variable has a value,
    its value is used for DIR.

Options:
  -L      force symbolic links to be followed: resolve symbolic
          links in DIR after processing instances of `..'
  -P      use the physical directory structure without following
          symbolic links: resolve symbolic links in DIR before
          processing instances of `..'
  -e      if the -P option is supplied, and the current working
          directory cannot be determined successfully, exit with
          a non-zero status
  -@      on systems that support it, present a file with extended
          attributes as a directory containing the file attributes

The default is to follow symbolic links, as if `-L' were specified.
`..' is processed by removing the immediately previous pathname component
back to a slash or the beginning of DIR.

Exit Status:
Returns 0 if the directory is changed, and if $PWD is set successfully when
-P is used; non-zero otherwise.
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 10: Справка по команде cd


```
PWD(1)                                User Commands                                PWD(1)

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the version described here.
    Please refer to your shell's documentation for details about the options it supports.

AUTHOR
    Written by Jim Meyering.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

COPYRIGHT
    Manual page pwd(1) line 1 (press h for help or q to quit)
```

Рисунок 11: Справка по команде pwd

```
MKDIR(1)                                User Commands                                MKDIR(1)

NAME
    mkdir - make directories

SYNOPSIS
    mkdir [OPTION]... DIRECTORY...

DESCRIPTION
    Create the DIRECTORY(ies), if they do not already exist.

    Mandatory arguments to long options are mandatory for short options too.

    -m, --mode=MODE
        set file mode (as in chmod), not a=rwx - umask

    -p, --parents
        no error if existing, make parent directories as needed, with their file modes unaffected by
        any -m option

    -v, --verbose
        print a message for each created directory

    -Z
        set SELinux security context of each created directory to the default type

    --context[=CTX]
        like -Z, or if CTX is specified then set the SELinux or SMACK security context to CTX

    --help
        display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Рисунок 12: Справка по команде mkdir

```
RMDIR(1)                                User Commands                                RMDIR(1)

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

REPORTING BUGS
    GNU coreutils online help: <https://www.gnu.org/software/coreutils/>
    Report any translation bugs to <https://translationproject.org/team/>

COPYRIGHT
    Copyright © 2025 Free Software Foundation, Inc.  License GPLv3+: GNU GPL version 3 or later
    <https://gnu.org/licenses/gpl.html>.
    Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Рисунок 13: Справка по команде rmdir

```
RM(1)                                User Commands                                RM(1)

NAME
  rm - remove files or directories

SYNOPSIS
  rm [OPTION]... [FILE]...

DESCRIPTION
  This manual page documents the GNU version of rm.  rm removes each specified file.  By default, it does not remove directories.

  If the -I or --interactive=once option is given, and there are more than three files or the -r, -R, or --recursive are given, then rm prompts the user for whether to proceed with the entire operation.  If the response is not affirmative, the entire command is aborted.

  Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is not given, or the -i or --interactive=always option is given, rm prompts the user for whether to remove the file.  If the response is not affirmative, the file is skipped.

OPTIONS
  Remove (unlink) the FILE(s).

  -f, --force
        ignore nonexistent files and arguments, never prompt

  -i
        prompt before every removal

  -I
        prompt once before removing more than three files, or when removing recursively; less intrusive than -i, while still giving protection against most mistakes

  --interactive[=WHEN]
        prompt according to WHEN: never, once (-I), or always (-i); without WHEN, prompt always

  --one-file-system

Manual page rm(1) line 1 (press h for help or q to quit)
```

Рисунок 14: Справка по команде rm

```
157  ls -l
158  ls -f
159  cd /var/spool/
160  ls -l
161  cd
162  ls
163  ls -al
164  mkdir newdir
165  mkdir newdir/morefun
166  mkdir letters memos misk
167  ls
168  rm letters/ memos/ misk/
169  rm -r letters/ memos/ misk/
170  rm -r newdir/
171  ls
172  ls -R
173  ls -t
174  help cd
175  man pwd
176  man mkdir
177  man rmdir
178  man rm
179  history
ramielsarhanov@ramielsarhanov:~$
```

Рисунок 15: Команда history

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия пользователя с системой посредством командной строки.